

Ali Khalid

📍 Rochester, NY

✉ alikhali3110@gmail.com

🌐 ali-khalid-6696a7167

🔗 alikhali31.github.io

Professional Summary

I develop efficient and scalable 3D data pipelines for autonomous driving and sports analytics. My work spans the entire 3D data lifecycle, encompassing sensing, processing, compression, and storage, with hands-on experience utilizing LiDAR and stereo cameras. I combine expertise in machine learning and 3D computer vision to design and implement end-to-end systems using ROS.

Education

2022 - 2027
Rochester, NY

PhD Computer Science, Rochester Institute of Technology
Dissertation: Efficient and Scalable 3D Perception under Resource Constraints.
Skills: 3D Computer Vision, Multi-view Geometry, Sensor Fusion

2016 - 2020
Lahore, Pakistan

BSc Electrical Engineering, University of Engineering and Technology
Thesis: A smart assistant for homes and offices.
Skills: Machine Learning, Embedded Systems, Robotics, Signal Processing

Recent Publications

- 2026 **Been There, Scanned That: Nostalgia-Driven LiDAR Compression for Self-Driving Cars**
Ali Khalid, Jaiaid Mobin, Sumanth Appala, Avinash Maurya, ..., Fawad Ahmad
ACM SenSys'26, Intl. Conference on Embedded Artificial Intelligence and Sensing Systems, 2026
- 2024 **VRF: Vehicle Road-side Point Cloud Fusion** | Paper | Code & Dataset
Ali Khalid*, Kaleem Khan*, Yash Turkar, Kartik Dantu, Fawad Ahmad (*coprimary authors)
ACM MobiSys'24, International Conference on Mobile Systems, 2024

Research Experience

2022 - present
Rochester, NY

Graduate Research Assistant, Rochester Institute of Technology

- Developing an AI camera system for real-time sports broadcasting, with capabilities in 3D ball tracking, play analysis, and camera angle optimization.
- Developed a CPU-based 3D LiDAR compression algorithm for autonomous vehicles by leveraging spatiotemporal redundancies over extended period achieving 3× better compression ratio with only 1.5× compute overhead compared to SOTA.
- Designed a real-time V2I cooperative perception system for autonomous vehicles, enabling 3D LiDAR data transmission, localization, and fusion from roadside units to vehicles with 20 ms latency and 2 cm accuracy, surpassing SOTA performance.

Skills: Sensor Fusion, LiDARs, Stereo Cameras, CARLA, ROS, PCL, Open3D

2025 - 2025
Murray Hill, NJ

Summer Research Intern, Nokia Bell Labs

- Designed and implemented the first predictive teledriving system leveraging motion forecasting to mitigate the impact of high network latency and ensure responsive remote vehicle control.

Skills: Multimodal Motion Prediction, GPU Scheduling, Rendering, CARLA

2021 - 2022
Lahore, Pakistan

Research Assistant, Lahore University of Management Sciences

- Spearheaded research in SAR polarimetry for enhanced land use classification accuracy and led efforts in green space identification, tree counting, travel time estimation, and road network extraction using satellite imagery.

Skills: 2D Object Detection & Segmentation, GEE, QGIS, ArcGIS, SNAP

2020 - 2021 Lahore, Pakistan	Research Assistant, Information Technology University Developed innovative techniques for image relocalization through scene coordinates regression, estimating occlusion-aware optical flow using deep learning, and generating detailed 3D models of indoor environments using Structure From Motion, enhancing spatial understanding and navigation capabilities. Skills: Structure From Motion, Optical Flow, COLMAP, OpenSFM
2019 - 2019 Lahore, Pakistan	Research Intern, Lahore University of Management Sciences Conducted research leveraging Wi-Fi CSI to develop solutions for estimating rotor speeds of industrial machines, passively tracking multiple individuals, estimating occupancy through walls, and detecting soil moisture levels. Skills: Wireless Localization and Tracking, USRP, MATLAB
2018 - 2018 Lahore, Pakistan	Research Intern, Al-Khwarizmi Institute of Computer Science Worked in a team to develop a brain-controlled wheelchair, contributing to the acquisition of brain signal training data, training machine learning model, and implementing an obstacle avoidance feature. Skills: Emotive Epoc+, MindWave, Ultrasonic Distance Sensor, Raspberry Pi

Teaching Experience

2025 Rochester, NY	Graduate Teaching Assistant, Rochester Institute of Technology Courses: Foundations of Computer vision
2018-2019 Lahore, Pakistan	Graduate Teaching Assistant, University of Engineering and Technology Courses: Introduction to Computing, Digital Systems, Basic Electrical Technology

Related Skills

3D Computer Vision Object Detection & Tracking, Semantic Segmentation, Registration, Localization, Mapping	● ● ● ● ●	Deep Learning Transformers, GANs, CNNs, RNNs, NeRF, MLPs, VAEs, FPNs, RPNs	● ● ● ● ●
2D Computer Vision Object Detection & Tracking, Semantic Segmentation, Depth Estimation, Image Relocalization	● ● ● ● ●	Machine Learning Regression, Classification, Clustering, Reinforcement Learning, Diffusion models	● ● ● ● ●
Frameworks ROS, CARLA, OpenSFM, COLMAP	● ● ● ● ●	Software Infrastructure Python, C++, CUDA, Docker, Bash	● ● ● ● ●
Programming Libraries PCL, Open3D, OpenCV, NumPy, Pandas, SciPy	● ● ● ● ●	Machine Learning Frameworks PyTorch, TensorFlow, Keras	● ● ● ● ●